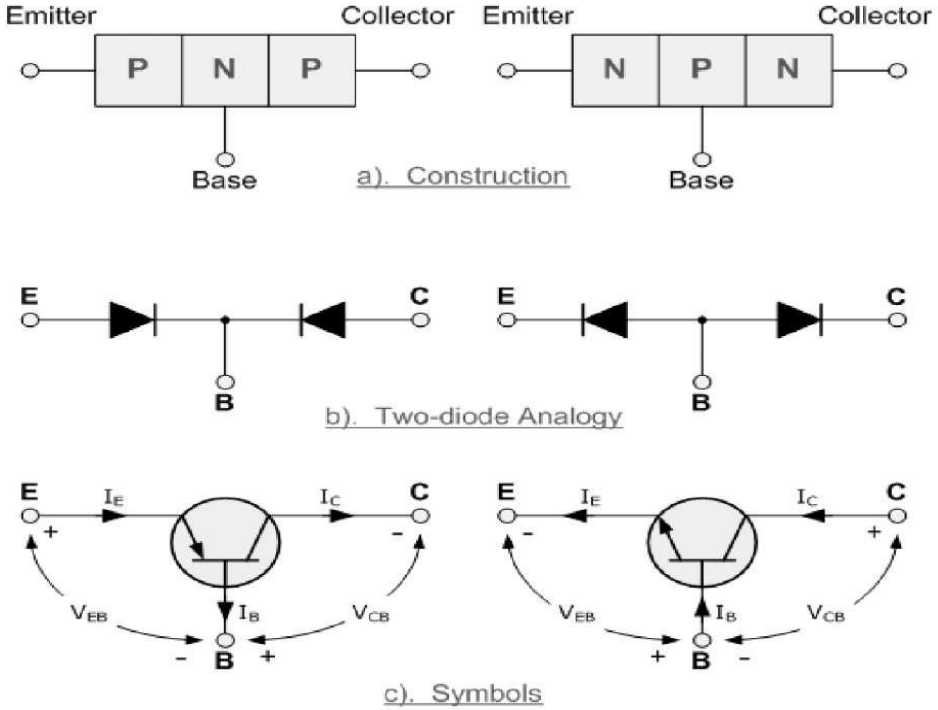
Bipolar Junction Transistors

Introduction

- The bipolar junction transistor (BJT) is a nonlinear, 3-terminal device.
- A transistor is a sandwich of one type of semiconductor (P-type or N-type) between two layers of other types.
- Current conduction in the transistor is based on the movement of two majority current carriers (holes and electrons).
- The term *bipolar* denotes this condition of conduction.
- The bipolar junction transistor is used in two broad areas of electronics as an amplifier to boost an electrical signal and as an electronic switch.

Introduction

- BJT Transistors are classified into two types:
- ***PNP* transistor:** It is obtained when an N-type layer of silicon is sandwiched between two P-type silicon materials. The arrowhead of the emitter lead "Points in" toward the base it indicates that the device is of the PNP type.
- ***NPN* transistor:** it is obtained when a p-type layer of silicon is sandwiched between two N- type silicon materials. An NPN transistor has the arrow "Not Pointing" toward the base.



Introduction

- The three portions of transistors are named as emitter, base and collector.
- The junction between emitter and base is called *emitter-base junction* while the junction between the collector and base is called *collector-base junction*.
- The base is thin and lightly doped. By making the base very thin and the impurity concentration in the base much less in the emitter and collector, free electrons (holes) emerging from the emitter have little chance of combining with the holes (electrons) in the base with the result that only about 98% of these electrons (holes) reach the collector.
- The emitter is heavily doped than any of the other regions because its main function is to supply majority charge carriers (either electrons or holes) to the base and it is wider when compared to the base.
- The width of the collector is more when compared to both the base and emitter to help dissipate heat that might be associated with the operation of the transistor.
- Its main function is to collect majority charge carriers coming from the emitter and passing through the base.

Introduction

- *In order to distinguish the emitter and collector an arrow is included in the emitter.*
- The arrowhead is always at the emitter (not at the collector) and in each case, its direction indicates the conventional direction of current flow when the E/B junction is forward biased.
- For a PNP transistor, arrowhead points from emitter to base meaning that emitter is positive with respect to base.
- For NPN transistor, it points from base to emitter meaning that base is positive with respect to the emitter.

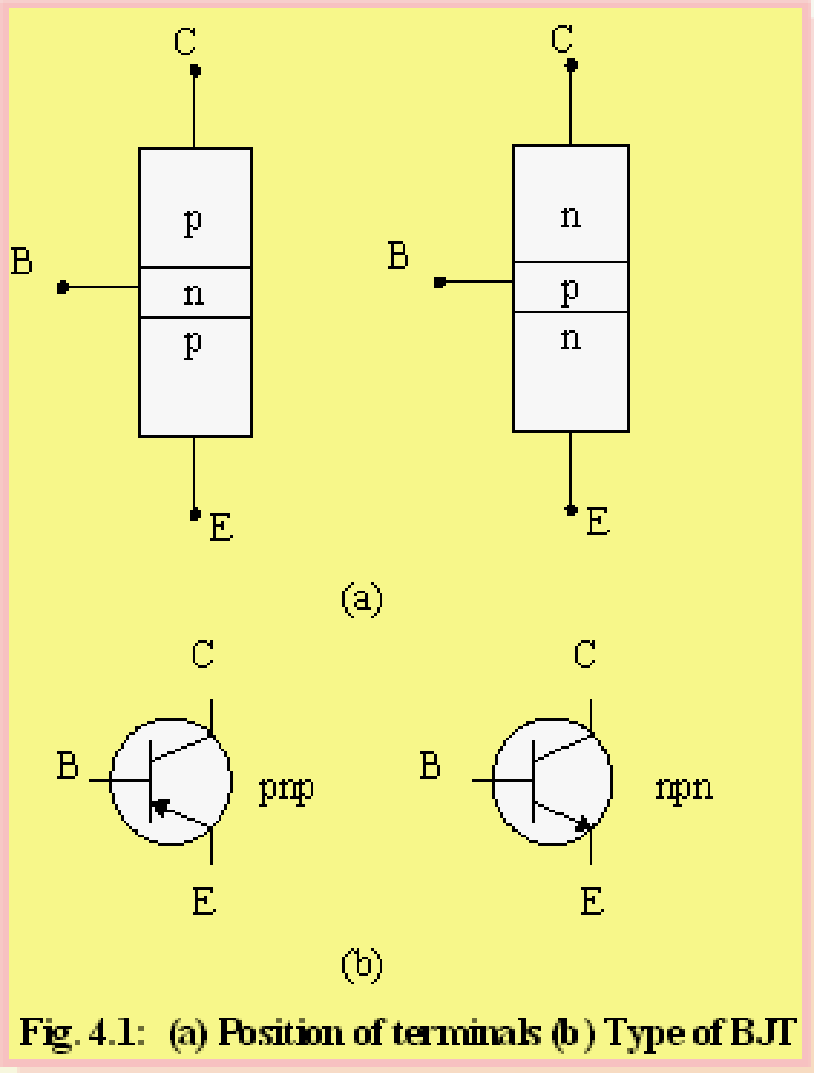
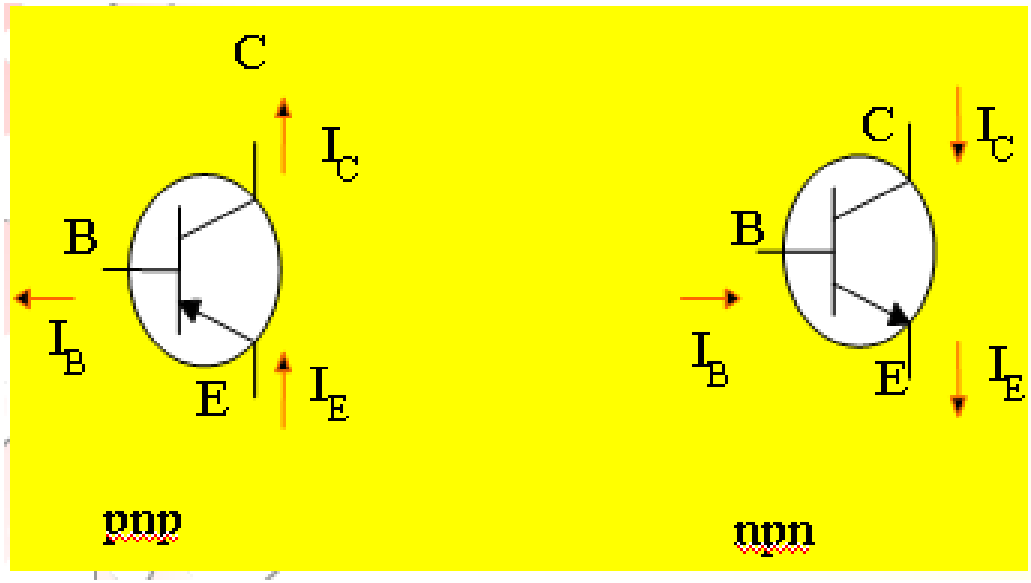


Fig. 4.1: (a) Position of terminals (b) Type of BJT

Transistor Currents

- There are three currents in a transistor:
  - Emitter Current, I_E
  - Collector Current, I_C
  - Base Current, I_B

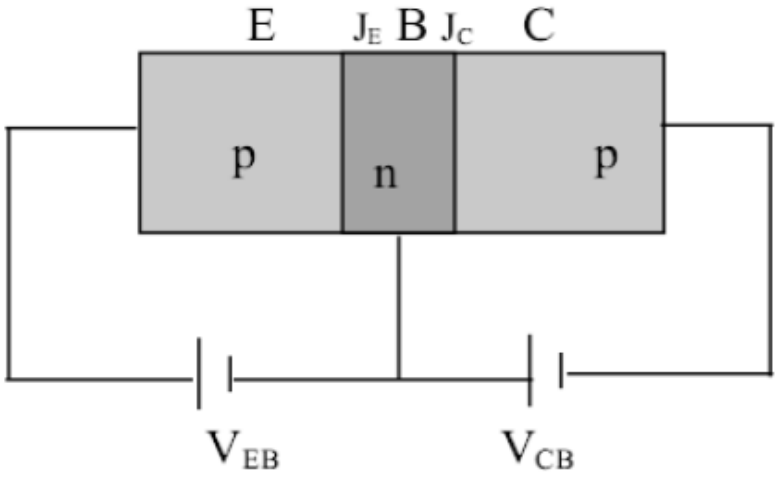


Operating Regions of a Transistor

- A transistor can be operated in three different regions as: *active region, saturation region and cut-off region*

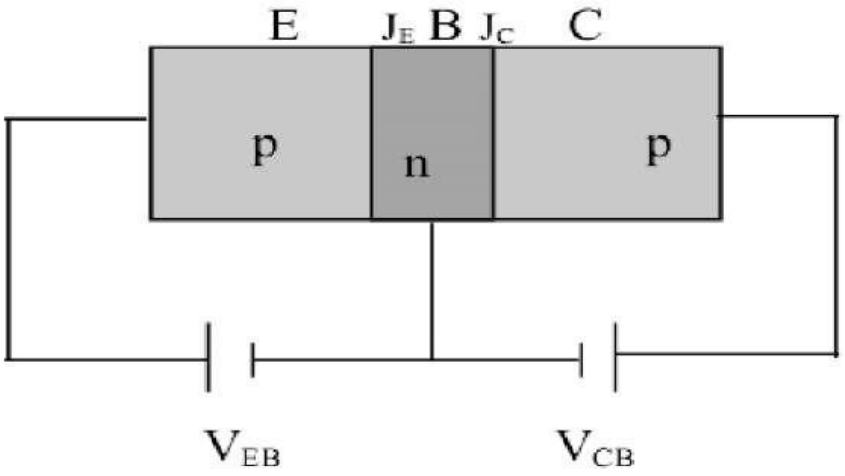
Active Region

- The transistor is said to be operated in active region when the emitter-base junction is forward biased and collector base junction is reverse biased as shown below.
- The collector current is said to have two current components one is due to the forward biasing of *EB* junction and the other is due to reverse biasing of *CB* junction.
- The collector current component due to the reverse biasing of the collector junction is called reverse saturation current and it is very small in magnitude.



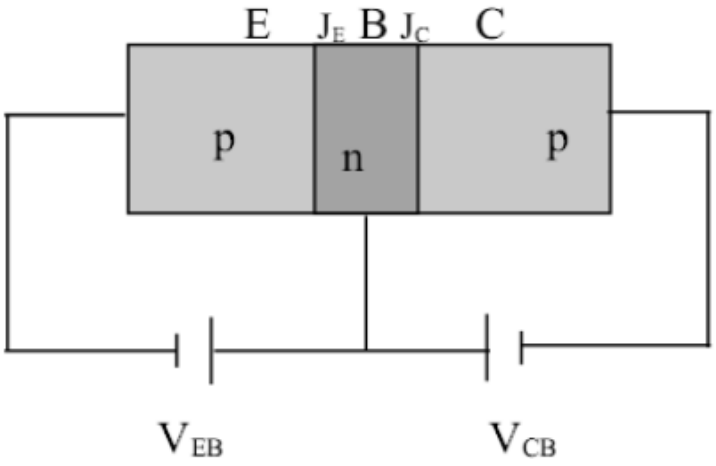
Saturation Region

- Transistor is said to be operating in saturation region when both *EB* junction and *CB* junction are forward biased as shown in the figure below
- When the transistor is operated in saturation region, I_C increases rapidly for a very small change in V_{CB}.
- *T*he transistor becomes saturated *i.e.* there is perfect short-circuit for both base-emitter and base-collector diodes.



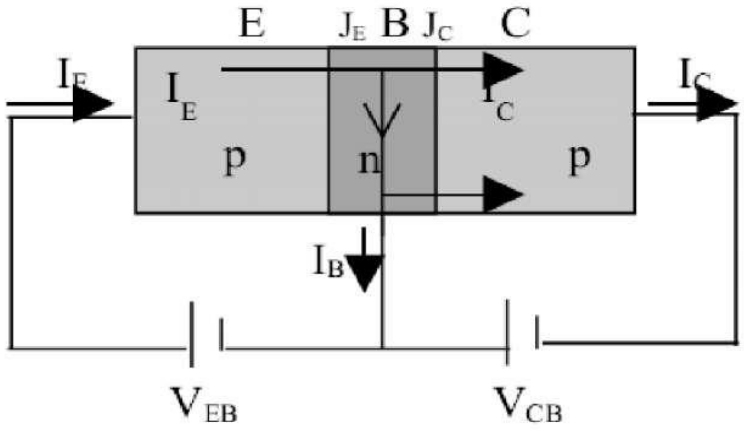
Cutoff Region

- When both *EB* junction (EBJ) and *CB* junction (CBJ) are reverse biased, the transistor is said to be operated in cut-off region as shown below
- In this region, the current in the transistor is very small and thus when a transistor in this region it is assumed to be in off state.



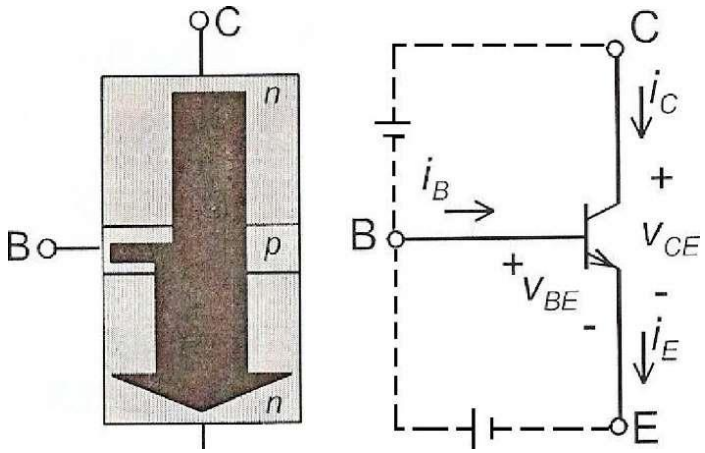
Theory of Operation of a Transistor (PNP)

- Consider a *PNP* transistor operated in active region as shown in figure below
- Since the EB junction is forward biased, large numbers of holes present in the emitter as majority carriers are repelled by the positive potential of the supply voltage V_{EB} and they move towards the base region causing emitter current I_E .
- Since the base is thin and lightly doped, very few of the holes coming from the emitter recombine with the electrons causing base current I_B and the entire remaining holes move towards the collector.
- Because the *CB* junction is reverse biased, all the holes are immediately attracted by the negative potential of the supply V_{CB}. Thereby giving rise to collector current, I_C.
- Thus: $\boldsymbol{I_E = I_B + I_C}$ (by ***KCL)***



Theory of Operation of a Transistor (NPN)

- Let us also consider the *NPN* transistor in active region (shown in figure below).
- Emitter region is heavily doped (lots of electrons available to conduct current).
- Emitter current, I_E, consists mostly of electrons being injected into base region; because the base is lightly doped, I_B is small.
- Some of the injected electrons combine with holes in base region. *Most* of the electrons travel across the narrow base and are attracted to the positive collector voltage, creating a collector current, I_C.
- The relative current magnitudes are indicated by the arrow thicknesses in the figure.



Biasing Rule

- For a *PNP* transistor, both collector (more negative) and base are negative with respect to the emitter.
- Similarly, for *NPN* transistor, both collector (more positive) and base are positive with respect to the emitter. The different potentials have been designated by double subscripts.
- The first subscript always represents the point or terminal which is more positive (or less negative) than the point or terminal represented by the second subscript.
- For example, in the figure below, the potential difference between emitter and base is written as V_{EB} (and not V_{BE}) because emitter is positive with respect to base.
- Between the base and collector themselves, collector is more negative than base. Hence, their potential difference is written as V_{BC} and not as V_{CB}.

